Top 23 CAT Remainders Questions With Video Solutions

Questions

Instructions

For the following questions answer them individually

Question 1

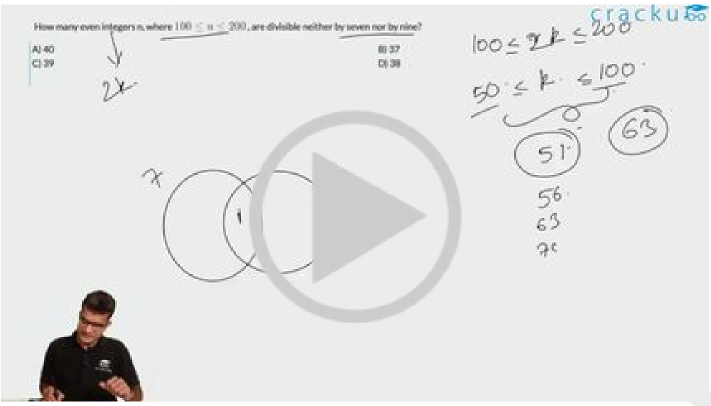
How many even integers n, where $100 \leq n \leq 200$, are divisible neither by seven nor by nine?

A 40

B 37

C 39

D 38



VIDEO SOLUTION

Question 2

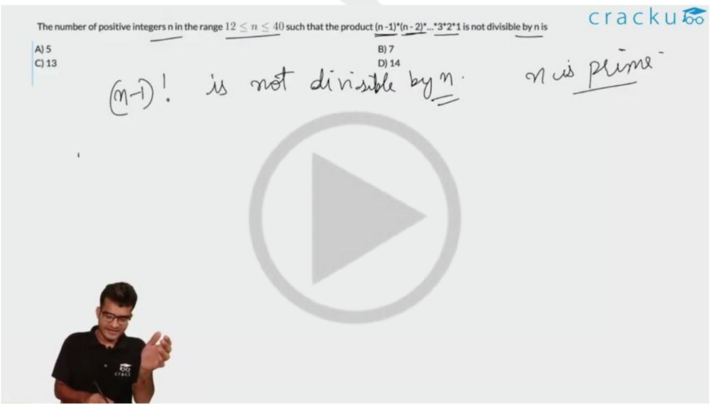
The number of positive integers n in the range $12 \leq n \leq 40$ such that the product (n -1)*(n - 2)*...*3*2*1 is not divisible by n is

A 5

B 7

C 13

D 14



VIDEO SOLUTION

Question 3

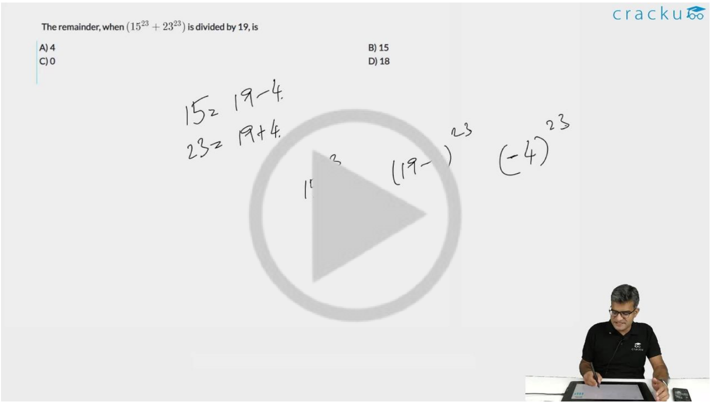
The remainder, when $(15^{23} + 23^{23})$ is divided by 19, is

A 4

B 15

C 0

D 18



VIDEO SOLUTION

Question 4

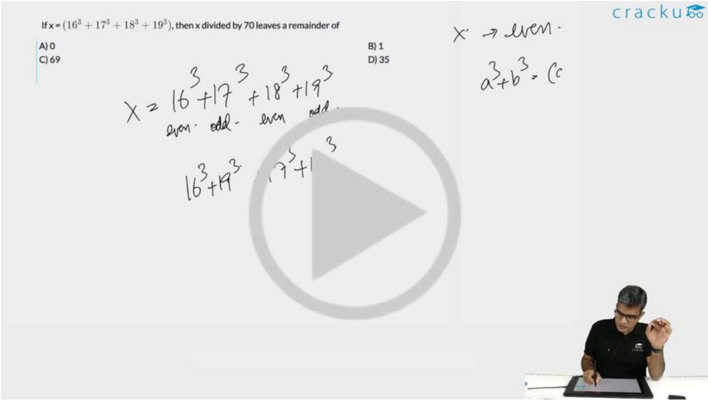
If x = $(16^3 + 17^3 + 18^3 + 19^3)$, then x divided by 70 leaves a remainder of

A 0

B 1

C 69

D 35



VIDEO SOLUTION

Question 5

Let $n! = 1 * 2 * 3 * ... * n$ for integer $n \geq 1$.

If $p = 1! + (2 * 2!) + (3 * 3!) + ... + (10 * 10!)$, then $p + 2$ when divided by 11! leaves a remainder of

A 10

B 0

C 7

D 1

VIDEO SOLUTION

Question 6

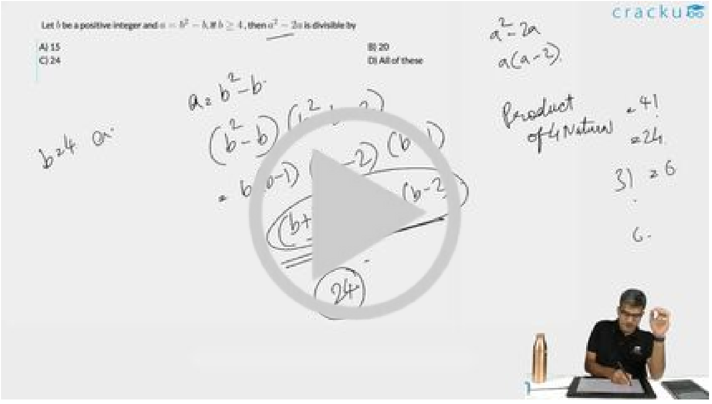
Let b be a positive integer and $a = b^2 - b$. If $b \geq 4$, then $a^2 - 2a$ is divisible by

A 15

B 20

C 24

D All of these



VIDEO SOLUTION

Question 7

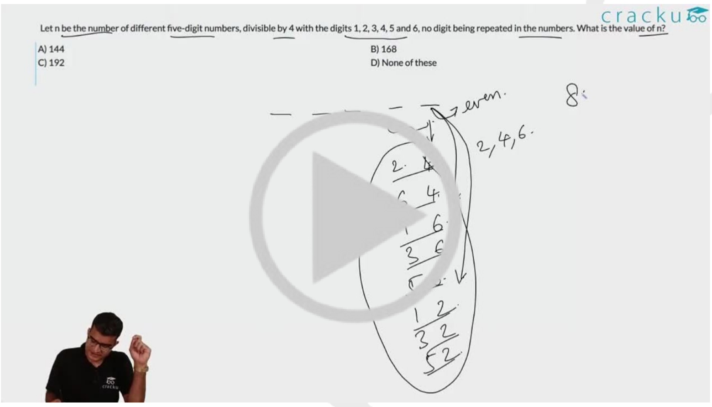
Let n be the number of different five-digit numbers, divisible by 4 with the digits 1, 2, 3, 4, 5 and 6, no digit being repeated in the numbers. What is the value of n?

A 144

B 168

C 192

D None of these



VIDEO SOLUTION

Question 8

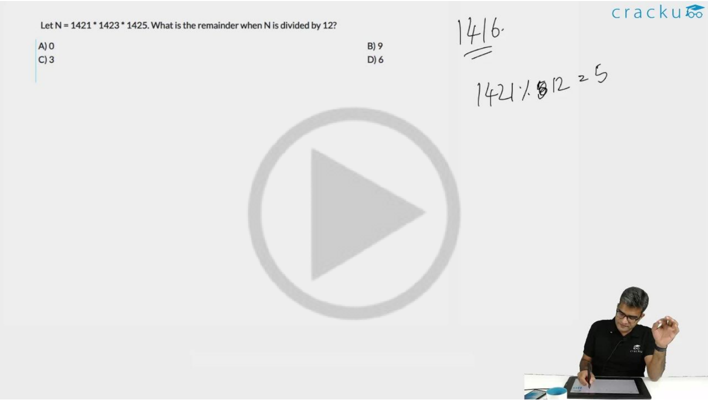
Let N = 1421 * 1423 * 1425. What is the remainder when N is divided by 12?

A 0

B 9

C 3

D 6



VIDEO SOLUTION

Question 9

The integers 34041 and 32506 when divided by a three-digit integer n leave the same remainder. What is n?

A 289

B 367

C 453

D 307

VIEW SOLUTION

Question 10

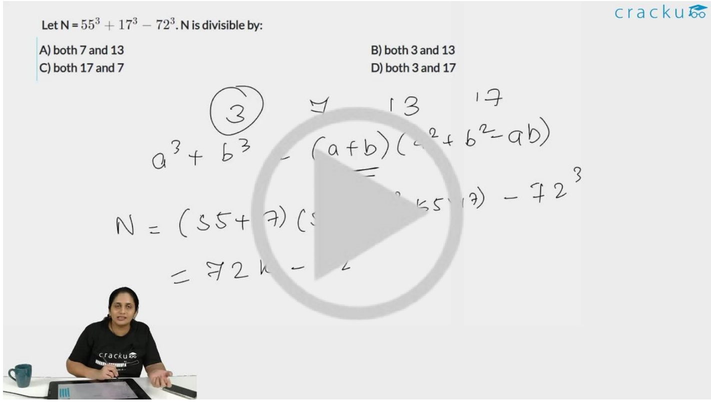
Let N = $55^3 + 17^3 - 72^3$. N is divisible by:

A both 7 and 13

B both 3 and 13

C both 17 and 7

D both 3 and 17



VIDEO SOLUTION

Question 11

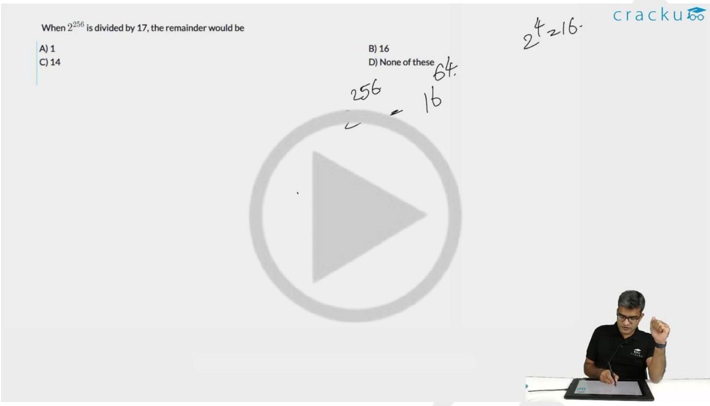
When 2^{256} is divided by 17, the remainder would be

A 1

B 16

C 14

D None of these



VIDEO SOLUTION

Question 12

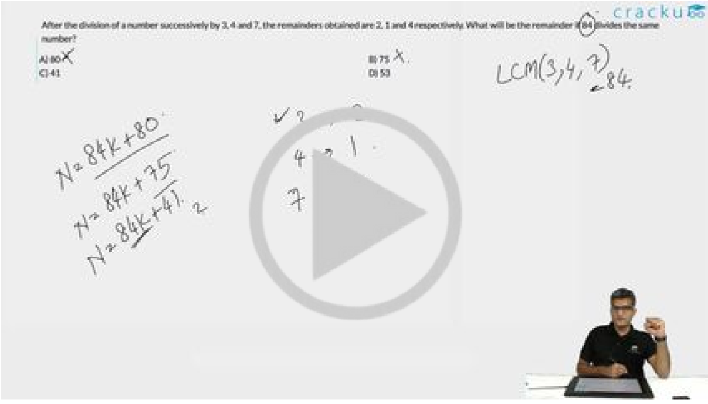
After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?

A 80

B 75

C 41

D 53



VIDEO SOLUTION

Question 13

$7^{6n} - 6^{6n}$, where n is an integer > 0, is divisible by

A 13

B 127

C 559

D All of these

VIEW SOLUTION

Question 14

The remainder when 2^{60} is divided by 5 equals

A 0

B 1

C 2

D None of these

VIEW SOLUTION

Question 15

Find the minimum integral value of n such that the division $\frac{55n}{124}$ leaves no remainder.

A 124

B 123

C 31

D 62

VIEW SOLUTION

Question 16

Let k be a positive integer such that k+4 is divisible by 7. Then the smallest positive integer n, greater than 2, such that k+2n is divisible by 7 equals

A 9

B 7

C 5

D 3

VIDEO SOLUTION

Question 17

A positive integer is said to be a prime number if it is not divisible by any positive integer other than itself and 1. Let p be a prime number greater than 5. Then $(p^2 - 1)$ is

A never divisible by 6

B always divisible by 6, and may or may not be divisible by 12.

C always divisible by 12, and may or may not be divisible by 24.

D always divisible by 24.

VIEW SOLUTION

Question 18

If m and n are integers divisible by 5, which of the following is not necessarily true?

A m - n is divisible by 5

B m^2 - n^2 is divisible by 25

C m + n is divisible by 10

D None of these

VIEW SOLUTION

Question 19

A certain number, when divided by 899, leaves a remainder 63. Find the remainder when the same number is divided by 29.

A 5

B 4

C 1

D Cannot be determined

VIEW SOLUTION

Question 20

A number is formed by writing first 54 natural numbers next to each other as 12345678910111213 ... Find the remainder when this number is divided by 8.

A 1

B 7

C 2

D 0

VIEW SOLUTION

Question 21

The remainder when 7^{84} is divided by 342 is :

A 0

B 1

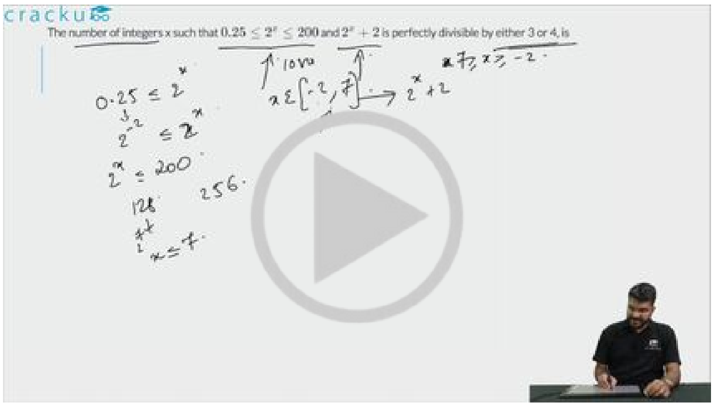
C 49

D 341

VIEW SOLUTION

Question 22

The number of integers x such that $0.25 \leq 2^x \leq 200$ and $2^x + 2$ is perfectly divisible by either 3 or 4, is



VIDEO SOLUTION

Question 23

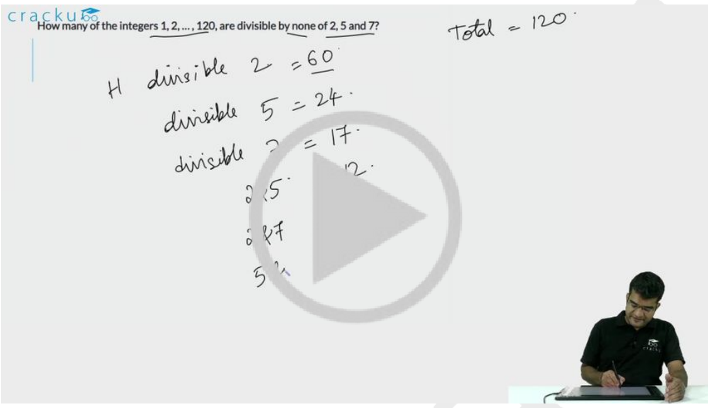
How many of the integers 1, 2, ... , 120, are divisible by none of 2, 5 and 7?

A 42

B 41

C 40

D 43



VIDEO SOLUTION

Answers

1.**C**	2.**B**	3.**C**	4.**A**	5.**D**	6.**C**	7.**C**	8.**C**
9.**D**	10.**D**	11.**A**	12.**D**	13.**D**	14.**B**	15.**A**	16.**A**
17.**D**	18.**C**	19.**A**	20.**C**	21.**B**	22.**5**	23.**B**	

Explanations

1.**C**

Between 100 and 200 both included there are 51 even nos. There are 7 even nos which are divisible by 7 and 6 nos which are divisible by 9 and 1 no divisible by both. hence in total 51 - (7+6-1) = 39

There is one more method through which we can find the answer. Since we have to find even numbers, consider the numbers which are divisible by 14, 18 and 126 between 100 and 200. These are 7, 6 and 1 respectively.

VIDEO SOLUTION

2.**B**

positive integers n in the range $12 \leq n \leq 40$ such that the product (n -1)*(n - 2)*...*3*2*1 is not divisible by n, implies that n should be a prime no. So there are 7 prime nos. in given range. Hence option B.

VIDEO SOLUTION

3.**C**

The remainder when 15^{23} is divided by 19 equals $(-4)^{23}$
The remainder when 23^{23} is divided by 19 equals 4^{23}
So, the sum of the two equals$(-4)^{23} + (4)^{23} = 0$

VIDEO SOLUTION

4.**A**

We know that x = $16^3 + 17^3 + 18^3 + 19^3 = (16^3 + 19^3) + (17^3 + 18^3)$

= $(16 + 19)(16^2 - 16 * 19 + 19^2) + (17 + 18)(17^2 - 17 * 18 + 18^2)$ = 35 × odd + 35 × odd = 35 × even = 35 × (2k)

=> x = 70k

=> Remainder when divided by 70 is 0.

VIDEO SOLUTION

5.**D**

According to given condiiton we have p = (1 × 1!) + (2 × 2!) + (3 × 3!) + (4 × 4!) + ... + (10 × 10!) . So n × n! = [(n + 1) - 1] × n! = (n + 1)! - n!. So equation becomes p = 2! - 1! + 3! - 2! + 4! - 3! + 5! - 4! +... + 11! - 10!. So p = 11! - 1! = 11! - 1. p + 2 = 11! + 1 .So when it is divided by 11! gives a remainder of 1. Hence, option 4.

VIDEO SOLUTION

6.**C**

We know that a=$b^2 - b$.

So$a^2 - a$ = b($b^3 - 2b^2 - b + 2$) . = (b - 2)(b - 1)(b)(b + 1)

The above given is a product of 4 consecutive numbers with the lowest number of the product being 2(given b >= 4)

In any set of four consecutive numbers, one of the numbers would be divisible by 3 and there would be two even numbers with the minimum value of the pair being (2,4).

Thus, for any value of b >=4, $a^2 - 4$ would be divisible by 3 x 2 x 4 = 24.

Thus, option C is the right choice. Options A and B are definitely wrong as a set of four consecutive numbers need not always include a multiple of 5 eg:(6,7,8,9)

VIDEO SOLUTION

7. **C**

To be divisible by 4 , last 2 digits of the 5 digit no. should be divisible by 4 . So possibilities are 12,16,32,64,24,36,52,56 which are 8 in number. Remaining 3 digits out of 4 can be selected in ${}^4C_3$ ways and further can be arranged in 3! ways . So in total = 8*4*6 = 192

VIDEO SOLUTION

8. **C**

The numbers 1421, 1423 and 1425 when divided by 12 give remainder 5, 7 and 9 respectively.

5*7*9 mod 12 = 11 * 9 mod 12 = 99 mod 12 = 3

VIDEO SOLUTION

9. **D**

The difference of the numbers = 34041 - 32506 = 1535

The number that divides both these numbers must be a factor of 1535.

307 is the only 3 digit integer that divides 1535.

VIEW SOLUTION

10. **D**

$55^3 + 17^3 - 72^3$ = $(55 - 72)k + 17^3$. This is divisible by 17

Remainder when 55^3 is divided by 3 = 1

Remainder when 17^3 is divided by 3 = -1

Remainder when 72^3 is divided by 3 = 0

So, $55^3 + 17^3 - 72^3$ is divisible by 3

So, the answer is d) 3 and 17

VIDEO SOLUTION

11. **A**

$2^4 = 16 = -1$ (mod 17)
So, $2^{256} = (-1)^{64}$(mod 17)
$= 1$ (mod 17)
Hence, the answer is 1. Option a).

VIDEO SOLUTION

12. **D**
Since after division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively, the number is of form ((((4*4)+1)*3)+2)k = 53K

Let k = 1; the number becomes 53

If it is divided by 84, the remainder is 53.

Option d) is the correct answer.

Alternative Solution.

Consider only for 3 and 4 and the remainders are 2 and 1 respectively.

So 5 is the first number to satisfy both the conditions. The number will be of the form 12k+5. Put different integral values of k to find whether it will leave remainder 5 when divided by 7. So the first number to satisfy such condition is 48x4+5= 53

VIDEO SOLUTION

13. **D**
Consider n=1 we have $7^6 - 6^6$ which is = $(7^3 + 6^3)(7^3 - 6^3)$ = 13 * 127 * 43 which is divisible by all the 3 options.

Option d) is the correct answer.

VIEW SOLUTION

14. **B**
2^{60} or 4^{30} when divided by 5
So according to remainder theorem
remainder will be $(-1)^{30}$ = 1.

VIEW SOLUTION

15. **A**
As 55 and 124 don't have any common factor, and n has to be a minimum integer, Hence, it should be 124 only. So that given equation won't have a remainder.

VIEW SOLUTION

16. **A**
let's say k+4 = 7m
k = 7m-4
Now for k+2n or 7m+(2n-4) is also multiple of 7.
or 2n-4 should be a multiple of 7

So 2n-4 = 7p
or 2n = 7p+4
For p=2; n=9 (p cannot be 1 as n is an integer)

VIDEO SOLUTION

17. **D**
Let the Prime number be 6n+1.

So $(p^2 - 1)$ = 6n(6n+2) = 12n(3n+1)

For any value of n , n(3n+1) will have a factor of 2

Hence given equation will be always be divisible by 24

VIEW SOLUTION

18. **C**
Let's say m=5k and n=5t
So m-n = 5(k-t) will be divisible by 5.
$m^2 - n^2 = 25(k^2 - t^2)$ will be divisible by 5.
$m + n = 5(k + t)$ will be divisible by 5 but not necessarily with 10.

VIEW SOLUTION

19. **A**
Let's say N is our number
N = (899K + 63) or N = $(29 \times 31$K) + 63
So when it is divided by 29, remainder will be $\frac{63}{29}$ = 5

VIEW SOLUTION

20. **C**
For a number to be divisible by 8, last 3 digits must be divisible by 8.

Last 3 digits of this number are 354.

354 mod 8 = 2

Hence, 2 is the remainder.

VIEW SOLUTION

21. **B**
7^3 = 343

7^{84} = $(7^3)^{28}$ = 343^{28}

343^{28} mod 342 = 1^{28} mod 342 = 1

VIEW SOLUTION

22. **5**

At $x = 0, 2^x = 1$ which is in the given range [0.25, 200]

$2^x + 2$ = 1 + 2 = 3 Which is divisible by 3. Hence, x = 0 is one possible solution.

At $x = 1, 2^x = 2$ which is in the given range [0.25, 200]

$2^x + 2$ = 2 + 2 = 3 Which is divisible by 4. Hence, x = 1 is one possible solution.

At $x = 2, 2^x = 4$ which is in the given range [0.25, 200]

$2^x + 2$ = 4 + 2 = 6 Which is divisible by 3. Hence, x = 2 is one possible solution.

At $x = 3, 2^x = 8$ which is in the given range [0.25, 200]

$2^x + 2$ = 8 + 2 = 3 Which is not divisible by 3 or 4. Hence, x = 3 can't be a solution.

At $x = 4, 2^x = 16$ which is in the given range [0.25, 200]

$2^x + 2$ = 16 + 2 = 18 Which is divisible by 3. Hence, x = 4 is one possible solution.

At $x = 5, 2^x = 32$ which is in the given range [0.25, 200]

$2^x + 2$ = 32 + 2 = 34 Which is not divisible by 3 or 4. Hence, x = 5 can't be a solution.

At $x = 6, 2^x = 64$ which is in the given range [0.25, 200]

$2^x + 2$ = 64 + 2 = 66 Which is divisible by 3. Hence, x = 6 is one possible solution.

At $x = 7, 2^x = 128$ which is in the given range [0.25, 200]

$2^x + 2$ = 128 + 2 = 130 Which is not divisible by 3 or 4. Hence, x = 7 can't be a solution.

At $x = 8, 2^x = 256$ which is not in the given range [0.25, 200]. Hence, x can't take any value greater than 7.

Therefore, all possible values of x = {0,1,2,4,6}. Hence, we can say that 'x' can take 5 different integer values.

VIDEO SOLUTION

23. **B**

The number of multiples of 2 between 1 and 120 = 60

The number of multiples of 5 between 1 and 120 which are not multiples of 2 = 12

The number of multiples of 7 between 1 and 120 which are not multiples of 2 and 5 = 7

Hence, number of the integers 1, 2, ... , 120, are divisible by none of 2, 5 and 7 = 120 - 60 - 12 - 7 = 41

VIDEO SOLUTION